# Supplementary material for "A continental-scale analysis of genomic variation in a songbird"

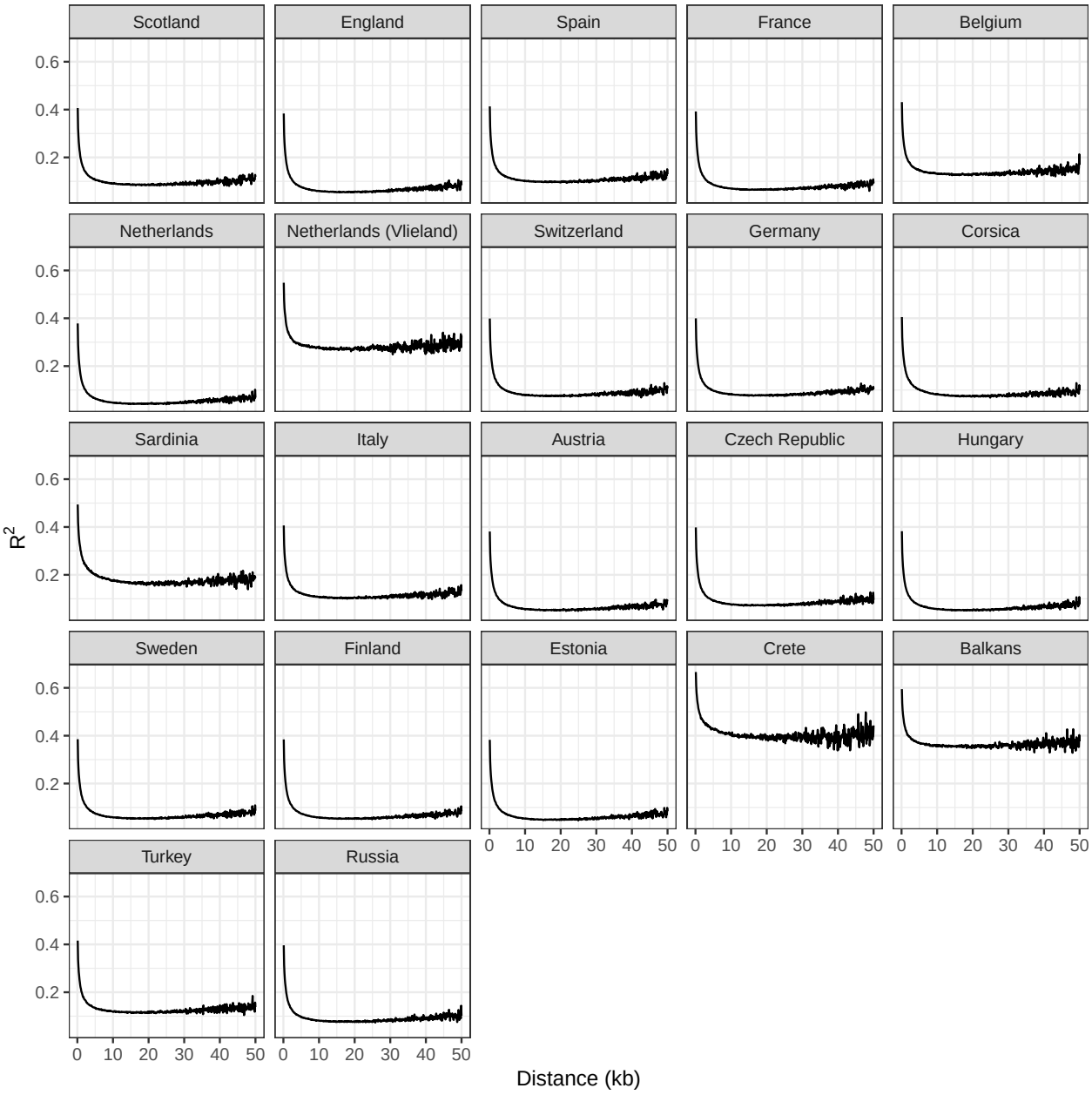


**Figure S1** Linkage disequilibrium in European great tit populations. Lines are means of $R^2$ values for all pairs of markers within 100bp distance bins.

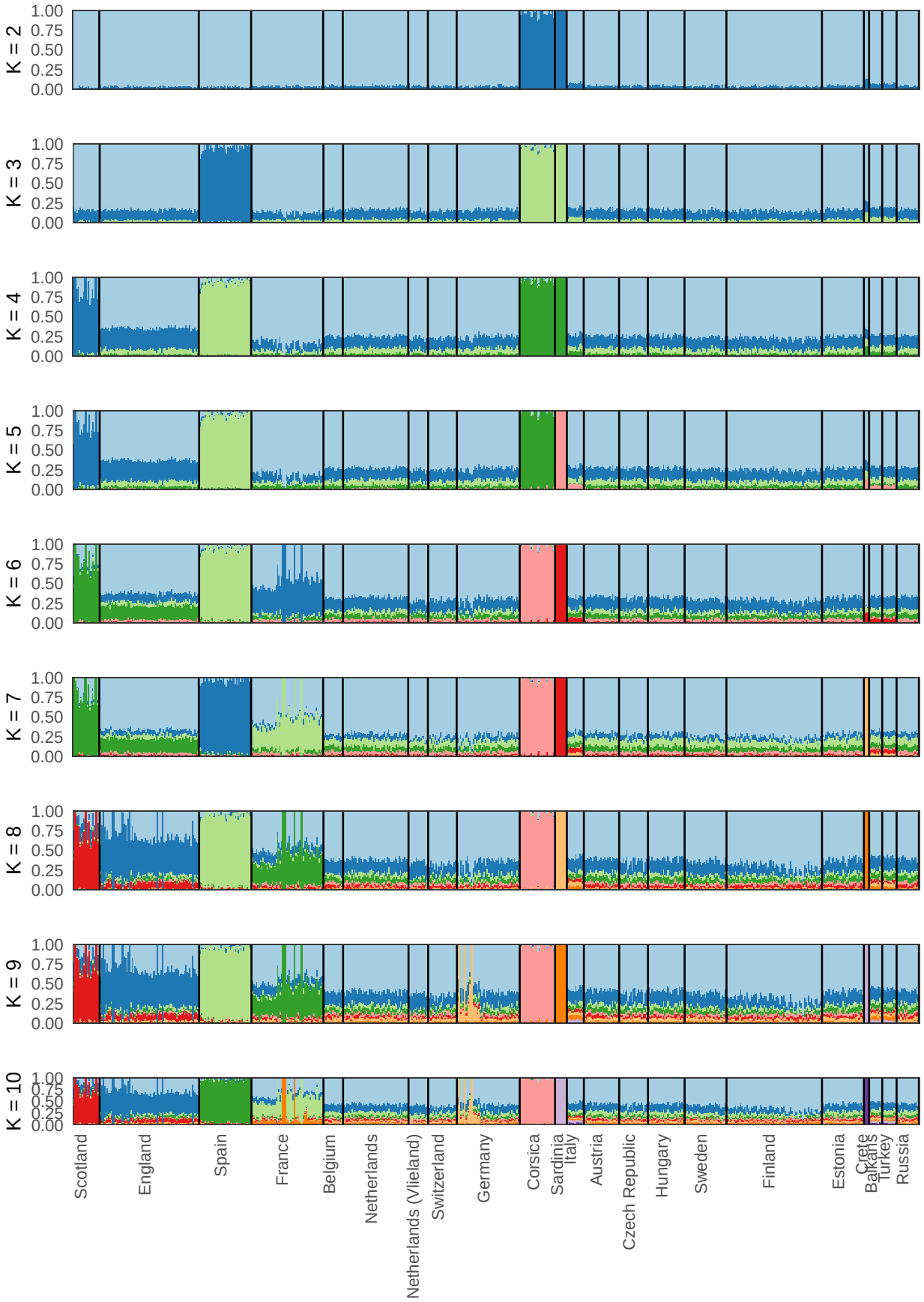
K = 2
K = 3
K = 4
K = 5
K = 6
K = 7
K = 8
K = 9
K = 10
1.00
0.75
0.50
0.25
0.00
Scotland
England
Spain
France
Belgium
Netherlands
Netherlands (Vlieland)
Switzerland
Germany
Corsica
Sardinia
Italy
Austria
Czech Republic
Hungary
Sweden
Finland
Estonia
Crete
Balkans
Turkey
Russia

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**Figure S2** Admixture analyses at $K = 2$ to 10 in European Great tit populations.

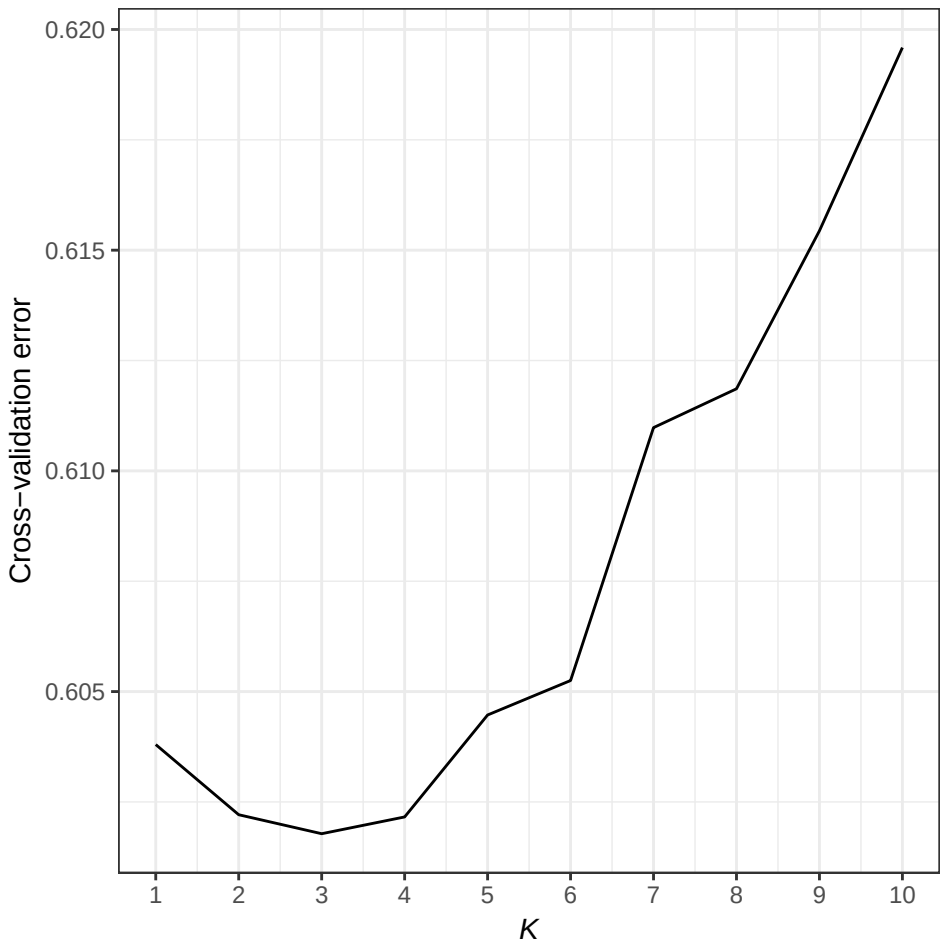


**Figure S3** Cross-validation error from admixture analyses at $K = 1$ to 10.

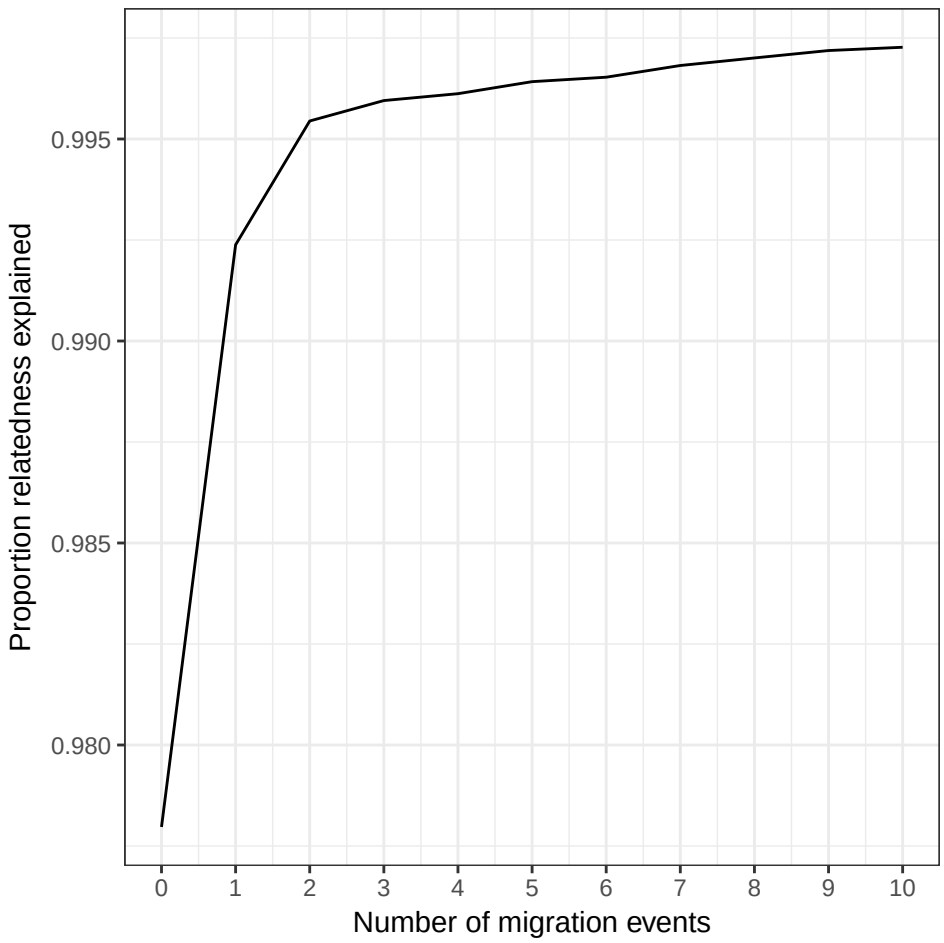


**Figure S4** Proportion of relatedness explained among populations in relation to the number of migration events allowed in Treemix analyses.

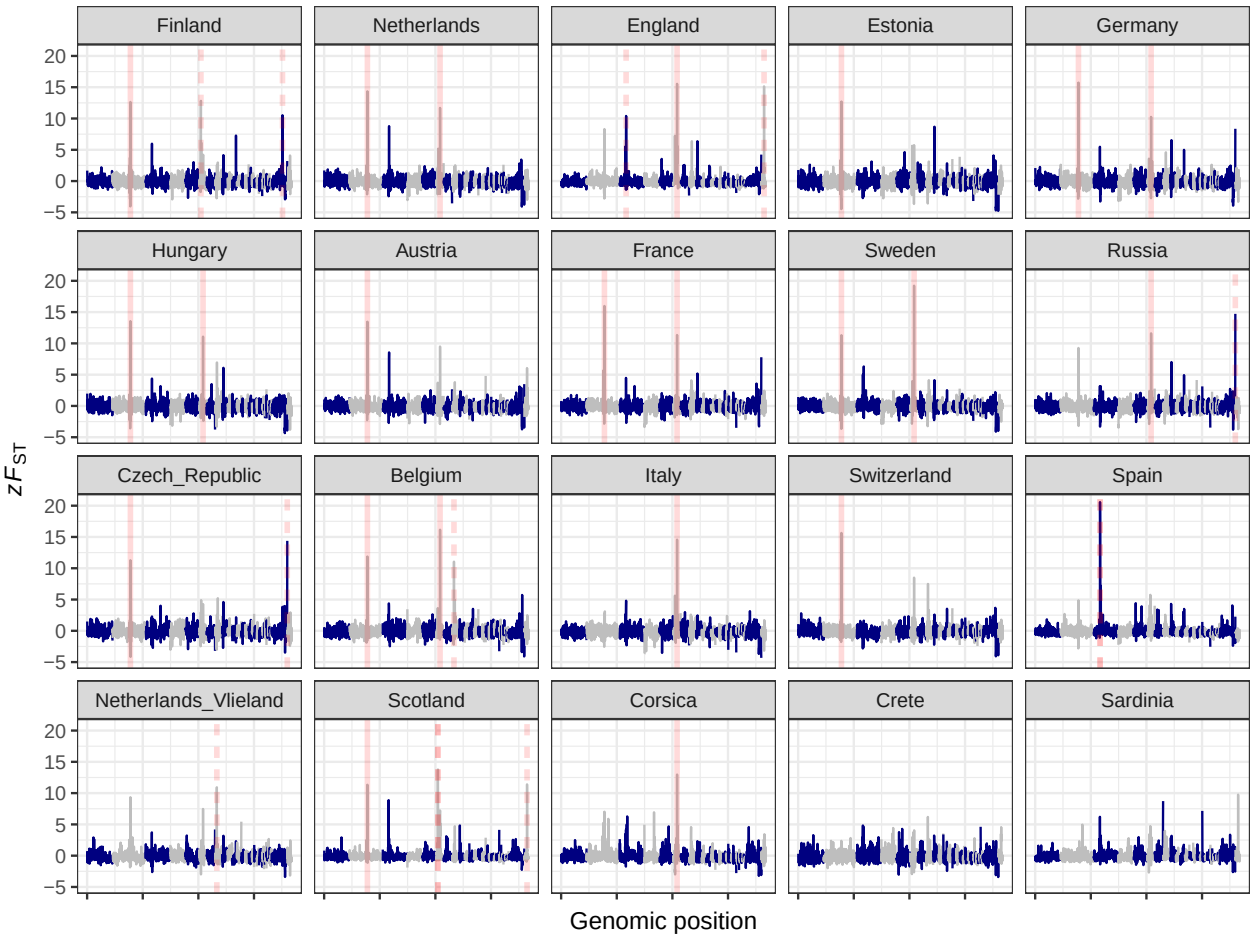


**Figure S5** Landscapes of relative genomic differentiation in European great tit populations. $zF_{ST}$ across the genome is averaged in 500kb windows, with each panel displaying a pairwise comparison between with the proposed refugial population in Turkey. Red lines represent $F_{ST}$ outliers (windows with mean $zF_{ST} > 10$) shared across more than two comparisons (solid red lines), or unique to one or two comparisons (dashed red lines).